

Erdős Problem #624

EP-624

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let X be a finite set of size n and $H(n)$ be such that there is a function $f : \{A : A \subseteq X\} \rightarrow X$ so that for every $Y \subseteq X$ with $|Y| \geq H(n)$ we have $\{f(A) : A \subseteq Y\} = X$. Prove that $H(n) - \log_2 n \rightarrow \infty$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>